| | | |
|---|---|---|
| | | Option D: Correct. Friction provided by turntable has a maximum value. Required centripetal force of Q will exceed maximum friction first due to larger *r*. |
| **9** | **C** | Horizontal: $T\ sin\ sin\ \theta\ = mr\omega^2 = mr\left(\frac{2\pi}{T}\right)^2 \ldots (1)$<br>Vertical: $T\ cos\ cos\ \theta\ = mg \ldots (2)$<br>(1)/(2): $tan\ tan\ \theta\ = \frac{4\pi^2 r}{gT^2} \Rightarrow T = \sqrt{\frac{4\pi^2 r}{gtantan\ \theta}}$ |
| **10** | **C** | Note that the satellites are *R* and 3*R* from the surface of Earth, not the centre of the Earth. |